Q1.1

$$x = \frac{r}{2}(\theta_l + \theta_r) \Rightarrow \dot{x} = \frac{r}{2}(\dot{\theta}_l + \dot{\theta}_r)$$

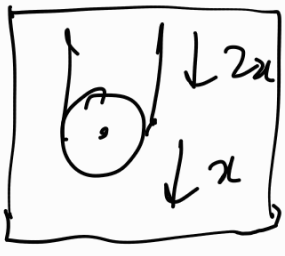
(r → motor radius)

$$y = \frac{r}{2}(\theta_l - \theta_r) \Rightarrow \dot{y} = \frac{r}{2}(\dot{\theta}_l - \dot{\theta}_r)$$

Both rotate same dir → change in x

Both rotate diff dir → change in y

$1/2$ due to belt



Q1.2

Friction Force $x = -\mu_{kx}\dot{x}$

Force $y = -\mu_{ky}\dot{y}$

$$\Rightarrow m\ddot{x} = F_x - b_x\dot{x}$$

$$m\ddot{y} = F_y - b_y\dot{y}$$

$$\Rightarrow \frac{mr}{2}(\dot{\omega}_l + \dot{\omega}_r) = F_x - \frac{b_x r}{2}(\omega_l + \omega_r)$$

$$\frac{mr}{2}(\dot{\omega}_l - \dot{\omega}_r) = F_y - \frac{b_y r}{2}(\omega_l - \omega_r)$$

For Motors

$$\mu = f\mu_r$$

$$\tau_{fl} = -\mu\omega_l$$

$$\tau_{fr} = -\mu\omega_r$$

$$J_m\dot{\omega}_l = K_m i_l - \mu\omega_l + \tau_l$$

$$= K_m i_l - \mu\omega_l - \frac{r}{2}[F_x + F_y]$$

$$\sim J_m\dot{\omega}_r = K_m i_r - \mu\omega_r - \frac{r}{2}[F_x - F_y]$$

Motor Constant:- $K_m \rightarrow \tau = K_m i_l$

$$\Delta V = K_m\dot{\theta}$$

$$z = [\theta_l \;\; \theta_r \;\; \dot{\theta}_l \;\; \dot{\theta}_r \;\; i_l \;\; i_r]^T$$

$$u = [v_l \;\; v_r]^T.$$

$$\frac{m_r}{2}(\dot{w}_l + \dot{w}_r) = F_x - \frac{b_{xr}}{2}(w_l + w_r)$$

$$\frac{m_r}{2}(\dot{w}_l - \dot{w}_r) = F_y - \frac{b_{yr}}{2}(w_l - w_r)$$

$$\Rightarrow F_x = \frac{m_r}{2}(\dot{w}_l + \dot{w}_r) + \frac{b_{xr}}{2}(w_l + w_r)$$

$$F_y = \frac{m_r}{2}(\dot{w}_l - \dot{w}_r) + \frac{b_{yr}}{2}(w_l - w_r)$$

$$\Rightarrow F_x + F_y = m_r \dot{w}_l + \left(\frac{b_{xr}}{2} + \frac{b_{yr}}{2}\right) w_l + \left(\frac{b_{xr}}{2} - \frac{b_{yr}}{2}\right) w_r$$

$$F_x - F_y = m_r \dot{w}_r + \left(\frac{b_{xr}}{2} - \frac{b_{yr}}{2}\right) w_l + \left(\frac{b_{xr}}{2} + \frac{b_{yr}}{2}\right) w_r$$

$$J_m \dot{\omega}_l = K_m i_l - \mu \omega_l - \frac{m s^2}{2} \dot{\omega}_l - \left(\frac{b_{xs} s^2}{4} + \frac{b_{ys} s^2}{4}\right) \omega_l - \left(\frac{b_{xs} s^2}{4} - \frac{b_{ys} s^2}{4}\right) \omega_r$$

$$J_m \dot{\omega}_r = K_m i_r - \mu \omega_r - \frac{m s^2}{2} \dot{\omega}_r - \left(\frac{b_{xs} s^2}{4} - \frac{b_{ys} s^2}{4}\right) \omega_l - \left(\frac{b_{xs} s^2}{4} + \frac{b_{ys} s^2}{4}\right) \omega_r$$

$$\dot{\theta}_l = \omega_l$$

$$\dot{\theta}_r = \omega_r$$

$$\dot{i}_l = \frac{u_l - R_m i_l - K_m \omega_l}{L_m}$$

$$\dot{i}_r = \frac{u_r - R_m i_r - K_m \omega_r}{L_m}$$

$$\Rightarrow \frac{\dot{i}_l + \dot{i}_r}{2}$$

$$= \frac{\overbrace{u_l + u_r}^{u_x}}{2} - R_m \left(\frac{\overbrace{i_l + i_r}^{i_x}}{2}\right) - \frac{K_m \left(\frac{\omega_l + \omega_r}{2}\right)}{L_m}$$

$$\dot{z} = Az + Bu \qquad z = [\theta_l \;\; \theta_r \;\; \omega_l \;\; \omega_r \;\; i_l \;\; i_r]^T$$

$$u = [v_l \;\; v_r]^T$$

$$A = \begin{bmatrix}
0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & P & Q & K_m' & 0 \\
0 & 0 & Q = R & P & 0 & K_m' \\
0 & 0 & -K_m/L_m & 0 & -R_m/L_m & 0 \\
0 & 0 & 0 & -K_m/L_m & 0 & -R_m/L_m
\end{bmatrix}$$

where

$$P = \frac{\left(-\mu + \frac{b_{xr}^2}{4} - \frac{b_{yr}^2}{4}\right)}{\left(J_m + \frac{m r^2}{2}\right)}$$

$$Q = \frac{\left(\frac{-b_{xr}^2}{4} + \frac{b_{yr}^2}{4}\right)}{\left(J_m + \frac{m r^2}{2}\right)}$$

$$K_m' = \frac{K_m}{\left(J_m + \frac{m r^2}{2}\right)}$$

$$S = P$$

$$B = \begin{bmatrix}
0 & 0 & 0 & 0 & 1/L_m & 0 \\
0 & 0 & 0 & 0 & 0 & 1/L_m
\end{bmatrix}^T$$

Q1.3

$$x = \frac{r}{2}(\theta_l + \theta_r) \quad \Rightarrow \quad \dot{x} = \frac{r}{2}(\dot{\theta}_l + \dot{\theta}_r)$$

$$y = \frac{r}{2}(\theta_l - \theta_r) \quad \Rightarrow \quad \dot{y} = \frac{r}{2}(\dot{\theta}_l - \dot{\theta}_r)$$

$$\ddot{x} = \frac{r}{2}(\ddot{\theta}_l + \ddot{\theta}_r) \qquad \ddot{y} = \frac{r}{2}(\ddot{\theta}_l - \ddot{\theta}_r)$$

— × —

For $\ddot{x}$:-

$$\frac{r}{2}(\ddot{\theta}_l + \ddot{\theta}_r) = \frac{r}{2}\left[P\omega_l + Q\omega_r + K_m i_l + R\omega_l + S\omega_r + K_m i_r\right]$$

$$= \frac{r}{2}\left[(P+Q)\omega_l + (Q+P)\omega_r + K_m'(i_l + i_r)\right]$$

For $\ddot{y}$:-

$$\frac{r}{2}(\ddot{\theta}_l - \ddot{\theta}_r) = \frac{r}{2}\left[P\omega_l + Q\omega_r + K_m' i_l - R\omega_l - S\omega_r + K_m' i_r\right]$$

$$= \frac{r}{2}\left[(P-Q)\omega_l + (Q-P)\omega_r + K_m'(i_l - i_r)\right]$$

For x:-

$$\dot{x} = x$$

$$\ddot{x} = \frac{r}{2}\left[P\omega_l + Q\omega_r + Km i_l + Q\omega_l + P\omega_r + Km i_r\right]$$

$$= \frac{r}{2}\left[(P+Q)\omega_l + (P+Q)\omega_r + Km'(i_l + i_r)\right]$$

$$= (P+Q)\left[\frac{r}{2}(\omega_l + \omega_r)\right] + \frac{Km' r}{2}(i_l + i_r)$$

$$= (P+Q)\dot{x} + Km' r\, i_x \qquad \left(i_x = \frac{i_l + i_r}{2}\right)$$

For y:-

$$\dot{y} = \dot{y}$$

$$\ddot{y} = (P-Q)\dot{y} + Km' r\, i_y \qquad \left(i_y = \frac{i_l - i_r}{2}\right)$$

$$\dot{i}_x = \frac{v_x - Rm\, i_x - Km\, r/r}{Lm} \qquad \dot{i}_y = \frac{v_y - Rm\, i_y - Km\, i_y/r}{Lm}$$

For x:-

$$\dot{x} = x$$

$$\ddot{x} = \frac{r K_m}{J_m} i_x$$

$$\dot{i}_x = \frac{v_x - R_m i_x - K_m \dot{\theta}}{L_m} \rightarrow \dot{x}/r$$

For y:-

$$\dot{y} = y$$

$$\ddot{y} = \frac{r K_m}{J_m} i_y$$

$$\dot{i}_y = \frac{v_y - R_m i_y - K_m \dot{\theta}}{L_m} \rightarrow \dot{y}/r$$

State Space for x_x

$$\dot{x}_x = A_x x_x + B_x v_x$$

$$A_x = \begin{bmatrix} 0 & 1 & 0 \\ 0 & (P+Q) & r K_m \\ 0 & -K_m/r L_m & -R_m/L_m \end{bmatrix} \qquad B_x = \begin{bmatrix} 0 \\ 0 \\ 1/L_m \end{bmatrix}$$

State Space for x_y

$$\dot{x}_y = A_y x_y + B_y x_y$$

$$A_y = \begin{bmatrix} 0 & 1 & 0 \\ 0 & (P-Q) & \gamma K_m' \\ 0 & -K_m/\gamma L_m & -R_m/L_m \end{bmatrix} \qquad B_y = \begin{bmatrix} 0 \\ 0 \\ 1/L_m \end{bmatrix}$$

Q 1.4

Simulation (will come to it later)

Q 1.5

$$v_n = L_n \dot{i}_n + V_n.$$

$$i_n = i_C + i_R$$

$$i_n = \frac{v_n - V_n}{L_n}$$

$$i_C = C_n \frac{dv_n}{dt} \qquad i_R = \frac{v_n}{R_n}$$

$$i_n = C_n \frac{dv_n}{dt} + \frac{v_n}{R_n}$$

$$\Rightarrow \dot{V}_n = \frac{1}{C_n}\left(i_n - \frac{V_n}{R_n}\right)$$

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for heating:- $P = \frac{V_n^2}{R_n} = i_R^2 R_n.$

$$C_{t/n}\,\dot{T}_n = P_n - \dot{Q}_{loss}$$

$$\dot{Q}_{loss} = \frac{T_n - T_e}{R_{t/n}}$$

$$\Rightarrow C_{t/n}\,\dot{T}_n = \frac{V_n^2}{R_n} - \frac{T_n - T_e}{R_{t/n}}$$

$$\Rightarrow \dot{T}_n = \frac{1}{C_{t/n}}\left[\frac{V_n^2}{R_n} - \frac{T_n - T_e}{R_{t/n}}\right]$$

State space $x_n = \begin{bmatrix} T_n \\ V_n \\ i_n \end{bmatrix}$ — V_n: V across C & R; i_n: curr across L

$$\dot{x}_n = f(x_n)$$

$$\dot{x}_n = \begin{bmatrix} \dfrac{V_n^2/R_n - (T_n - T_e)/R_{thn}}{C_{thn}} \\ \dfrac{i_n - V_n/R_n}{C_n} \\ \dfrac{u_n - V_n}{L_n} \end{bmatrix}$$

u_n: input.

Q 1.4

$$x_x = [x \quad \dot{x} \quad i_x]^T$$

$$\dot{x}_x = A_x x_x + B_x v_x$$

$$A_x = \begin{bmatrix} 0 & 1 & 0 \\ 0 & (\beta+\alpha) & \eta K_m' \\ 0 & -K_m/\eta L_m & -R_m/L_m \end{bmatrix} \qquad B_x = \begin{bmatrix} 0 \\ 0 \\ 1/L_m \end{bmatrix}$$

$$C_x = [1 \; 0 \; 0] \qquad D_x = 0$$

$$\dot{x}_y = A_y x_y + B_y x_y \qquad x_y = [y \; \dot{y} \; i_y]^T$$

$$A_y = \begin{bmatrix} 0 & 1 & 0 \\ 0 & (\beta-\alpha) & \eta K_m' \\ 0 & -K_m/\eta L_m & -R_m/L_m \end{bmatrix} \qquad B_y = \begin{bmatrix} 0 \\ 0 \\ 1/L_m \end{bmatrix}$$

$$C_y = [1 \; 0 \; 0] \qquad D_y = 0$$

Circle:- $x(t) = x_c + R\cos(\omega t)$

$y(t) = y_c + R\sin(\omega t)$

For x

$$\dot{i}_x = \frac{v_x - R_m i_x - K_m \dot{\theta}}{L_m}$$

$\dot{\theta} \rightarrow \dot{x}/r$

$$\therefore \dot{i}_x = \frac{v_x - R_m i_x - \frac{K_m \dot{x}}{r}}{L_m}$$

$$\ddot{x} = (P+Q)\dot{x} + r K_m' i_x$$

$$i_x = \frac{\ddot{x} - (P+Q)\dot{x}}{r K_m'} \quad \Big| \quad \dot{i}_x = \frac{\dddot{x} - (P+Q)\ddot{x}}{r K_m'}$$

$$\frac{\dddot{x} - (P+Q)\ddot{x}}{r K_m'} = \frac{v_x - R_m\left(\frac{\ddot{x} - (P+Q)\dot{x}}{r K_m'}\right) - \frac{K_m \dot{x}}{r}}{L_m}$$

$$\frac{L_m \ddot{x} - (P+Q) L_m \dot{x}}{r K_m'} = v_x - \frac{R_m \ddot{x}}{r K_m'} + \frac{R_m (P+Q)\dot{x}}{r K_m'} - \frac{K_m \dot{x}}{r}$$

$$\Rightarrow \boxed{U_x = \frac{L_m}{\eta K_m'}\dddot{x} + \left(\frac{R_m - (P+Q)L_m}{\eta K_m'}\right)\ddot{x} + \left(\frac{R_m(P+Q)}{\eta K_m'} - \frac{K_m}{\eta_g}\right)\dot{x}}$$